

90%

ABERDARE TECHNICAL COLLEGE
INFORMATION COMMUNICATION TECHNOLOGY
UNIT: PERFORM PROGRAMMING PRINCIPLES
UNIT CODE: 0613 451 05A

CAT 1
MODULE 4, TERM 4
TOTAL: 50 MARKS
TIME 1 HOURS 30 MIN

NAME BRIAN MAINA CDACC REG.....

DATE SIGN.....

Instructions

- ✓ This paper consists of **TEN (10)** questions and carries a total of **50 marks**.
- ✓ Answer **ALL** questions in the spaces provided.
- ✓ The time allowed for this paper is **1 hour**.
- ✓ Write your **Name, Admission/Registration Number** clearly on this sheet before you begin.
- ✓ Marks allocated to each question are shown in brackets at the end of the question or sub-question.
- ✓ Read each question carefully before answering. Where an example is required, ensure it is relevant and correctly stated.

QUESTIONS

1. a) Define the term "programming language." (2 marks)
- b) Programming languages can be grouped into different categories. Briefly describe **three** of the following categories, giving one example language for each: (5 marks)
 - Procedural languages
 - Object-oriented languages
 - Functional languages
2. Distinguish between **low-level** and **high-level** programming languages, giving one example of each. (4 marks)
3. State **four** criteria that should be considered when selecting a programming language to meet a given user's requirements (4 marks)
4. A hospital wants to develop a system for managing patient records, appointments, and billing. As the systems developer, identify and explain **five** criteria you would use to select an appropriate programming language for this system. (10 marks)
(Relate each criterion directly to the scenario.)
5. a) Define the term "programming paradigm." (2 marks)

→ Programming Language → Formal Language used by programmers to write instructions that a computer can understand and execute to perform specific tasks.

① Procedural languages → These organize a program into a sequence of procedures or functions that perform specific tasks. E.g. C

② Object-Oriented languages → These organize programs around objects and classes, which combine data & the operations performed on the data. E.g. Java

③ Functional languages → These treat computation mainly as the evaluation of functions and generally emphasize avoiding changes to data/state.

④ Low Level Language

- ① More difficult to maintain and write
- ② Usually hardware-dependent. E.g. Assembly language

High Level Language

- Easier to write, debug and maintain
- Generally more portable between systems. E.g. Python.

⑤ Application Requirements

- Ease of learning and use
- Performance
- Availability of tools and support.

70% Functional Programming → it solves problems by using functions and expressions, with emphasis on producing results from inputs rather than changing program state.

I Functional Programming → is a programming paradigm in which programs are constructed mainly using functions and computation is treated as the evaluation of those functions.

- ①
- Immutability
 - ~~Immutability~~
 - Use of pure functions.

8a) ^{Procedural} ~~Procedural~~ Programming → Programming ^{Paradigm} ~~Lamp~~ that organizes a program into a sequence of procedures or functions that operate on data to solve a problem.

- ② Advts
- Easy to divide into ^{Functions} ~~Procs~~ | Procedures.
 - Easy to understand.

9a Object - Oriented Programming → is a programming paradigm that organizes software around objects and classes, where objects contain data & methods that operate on that data.

b) Inheritance
Encapsulation.

10 Imperative Programming
Specifies how a problem should be solved.

uses instructions, statements and changes in program state.

Declarative Programming.
Specifies what result is required.

Describes the desired result without necessarily specifying the detailed steps.

1] Security $\Rightarrow$ The language should support security features because the system will store Confidential patient records and Billing information.

2] Database Support $\Rightarrow$ The language should have good support for database so that patient records appointments and bills can be retrieved efficiently.

3] Reliability $\Rightarrow$ Should allow developers to build a stable ~~database~~ ^{system} that minimizes errors & prevent loss or corrupt of patient information.

4] Performance $\Rightarrow$ Should provide good performance so that hospital staff can quickly retrieve patient records, book appointments & process bills.

5] Maintainability and Scalability $\Rightarrow$ The language should make it easy to ~~update~~ ^{update} the system and add features in the future. ~~8/10~~

6] Programming Paradigm $\Rightarrow$ is a style or approach used to structure and develop computer programs and solve problems. ~~8/10~~

6] Procedural programming $\Rightarrow$ "Solves a problem by breaking it into series of procedures or functions that are executed in a particular order."

- Object-oriented programming $\Rightarrow$ it solves problem by modelling them as objects and classes, where objects contain data and methods that operate on that data.